

Quantum Mechanics III

Homework 14

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Topic

Automated Instrument Calibration

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Automated Instrument Calibration. State all assumptions and conventions.

$$p(k|i) = \text{Tr}[E_k \mathcal{E}(\rho_i)]$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$J_{ai} = \partial p_a / \partial \theta_i$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$r_* = \text{argmin}_r(L_R, N_{fail}, d_{QDC})$$

Show enough intermediate work that another student can reproduce the calculation.